

KHORINSK, Russian FederationWMO: **307390**Lat: **52.167N**Lon: **109.785E**Elev: **666**StdP: **93.57**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.1	-36.7	-41.1	0.1	-38.8	-38.8	0.1	-36.6	8.4	-14.7	7.8	-15.6	0.2	50	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	30.1	18.4	28.3	17.6	26.4	16.8	19.9	27.3	18.8	25.5	17.9	24.0	2.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	13.6	22.2	16.6	12.8	21.4	15.5	12.0	21.0	60.2	27.4	56.5	25.5	53.3	24.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.4	DB	-41.1	33.7	3.6	2.8	-43.7	35.7	-45.8	37.4	-47.8	39.0	-50.4	41.0	(4)
(5)			WB	-40.7	21.8	3.0	1.7	-42.8	23.0	-44.6	23.9	-46.3	24.9	-48.4	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.1	-25.0	-19.9	-8.8	1.7	9.1	17.1	19.9	16.7	8.8	-0.9	-12.1	-21.4	(6)	
	DBStd	16.22	7.06	6.45	6.51	4.46	4.27	4.02	3.13	3.44	3.92	4.78	6.63	7.16	(7)	
	HDD10.0	4867	1083	837	582	249	69	2	0	1	68	338	662	974	(8)	
	HDD18.3	7225	1342	1070	840	498	287	70	19	72	286	596	912	1233	(9)	
	CDD10.0	807	0	0	0	1	41	214	307	209	33	1	0	0	(10)	
	CDD18.3	122	0	0	0	0	1	32	68	22	0	0	0	0	(11)	
	CDH23.3	1249	0	0	0	0	28	373	642	201	5	0	0	0	(12)	
	CDH26.7	345	0	0	0	0	3	99	208	35	0	0	0	0	(13)	
	Wind	WSAvg	2.4	1.9	1.7	2.4	3.1	3.2	2.9	2.4	2.3	2.5	2.4	1.9	(14)	
(15) Precipitation	PrecAvg	286	3	2	2	10	19	45	79	66	38	9	6	6	(15)	
	PrecMax	413	6	9	7	25	53	153	211	151	80	38	15	31	(16)	
	PrecMin	163	0	0	0	1	3	9	8	19	3	1	1	0	(17)	
	PrecStd	77	2	2	2	8	13	35	44	32	21	9	3	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.8	-1.8	8.7	18.3	27.1	32.0	33.4	29.8	23.8	14.6	4.8	-3.6	(19)	
		MCWB	-7.1	-4.3	3.0	8.9	14.8	19.4	20.1	18.6	14.3	8.1	0.7	-5.3	(20)	
	2%	DB	-8.6	-4.8	5.7	15.0	23.8	29.5	30.9	27.4	20.6	10.7	2.1	-5.7	(21)	
		MCWB	-9.8	-6.7	0.6	6.7	12.6	17.4	19.3	17.6	13.1	5.5	-0.8	-7.1	(22)	
	5%	DB	-11.8	-7.3	3.5	12.3	21.0	27.5	29.0	25.6	18.2	8.3	-0.3	-8.1	(23)	
		MCWB	-12.8	-8.8	-0.8	5.3	11.3	16.5	18.4	17.0	11.7	4.0	-2.4	-9.2	(24)	
	10%	DB	-14.6	-9.9	1.0	9.9	18.2	25.5	27.1	23.7	15.9	6.0	-2.4	-10.6	(25)	
		MCWB	-15.4	-11.1	-2.3	3.7	9.4	15.8	17.7	16.1	10.4	2.6	-4.2	-11.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.7	-4.1	3.1	9.2	15.2	20.5	22.0	20.2	15.8	8.6	1.0	-5.2	(27)	
		MCDB	-5.8	-2.0	8.5	17.6	25.0	30.4	29.9	26.4	21.4	13.6	4.1	-3.8	(28)	
	2%	WB	-9.7	-6.5	0.9	7.0	13.3	18.7	20.4	18.8	13.8	6.1	-0.8	-7.1	(29)	
		MCDB	-8.7	-4.8	5.2	14.1	22.9	26.6	28.4	24.4	19.4	10.1	1.9	-5.8	(30)	
	5%	WB	-12.9	-8.7	-0.7	5.6	11.6	17.6	19.4	17.8	12.4	4.3	-2.3	-9.2	(31)	
		MCDB	-11.9	-7.3	3.3	11.7	19.8	24.9	26.5	23.3	17.4	7.9	-0.4	-8.1	(32)	
	10%	WB	-15.4	-11.0	-2.2	4.0	10.1	16.5	18.5	16.9	10.8	2.7	-4.1	-11.5	(33)	
		MCDB	-14.6	-9.9	0.9	9.5	17.5	23.4	24.9	22.3	15.4	5.8	-2.5	-10.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	13.1	13.3	11.5	12.1	11.9	10.9	10.5	10.7	10.4	9.9	9.0	(35)	
		MCDBR	9.2	13.9	14.6	15.6	17.6	15.9	14.4	14.0	14.3	13.9	10.9	8.7	(36)	
	5% WB	MCWBR	8.5	12.3	10.2	9.8	9.7	7.2	5.7	6.2	7.8	9.0	8.4	7.7	(37)	
		MCWBR	9.2	13.7	14.2	14.6	15.9	13.4	12.1	11.1	13.1	13.2	10.8	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.256	0.281	0.313	0.363	0.450	0.445	0.412	0.374	0.332	0.300	0.279	0.259	(40)	
		taud	2.201	2.186	2.219	2.243	2.013	2.151	2.309	2.395	2.456	2.458	2.301	2.175	(41)	
	Ebn at Noon		745	832	876	870	802	811	836	852	854	815	714	663	(42)	
		Edh at Noon	67	92	110	123	163	144	121	105	87	71	61	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.28	2.33	3.79	5.04	5.72	6.00	5.54	4.69	3.59	2.27	1.28	0.93	(44)	
	RadStd		0.07	0.10	0.12	0.33	0.34	0.39	0.43	0.24	0.29	0.19	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	-1.75	-2.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page